

Supplementary Material

Multi-View 3D Wheat Ear Instance Segmentation for RGB-Based Phenotyping

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1 Multi-view 3D instance association

Algorithm 1 Multi-view 3D instance segmentation.

Require: 3DGS scene $\mathcal{G}$; selected views $\mathcal{V} = \{v_1, \dots, v_T\}$; precomputed 2D mask sets $\{\mathcal{M}_t\}_{t=1}^T$; SA3D-GS segmenter Φ ; matching threshold τ_{IoU}

Ensure: Predicted 3D instance set $\mathcal{O}$

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1:  $\mathcal{O} \leftarrow \emptyset$ 
2: for  $t = 1, \dots, T$  do
3:   Load  $\mathcal{M}_t = \{M_{t,k}\}_{k=1}^{K_t}$ 
4:   if  $\mathcal{O} = \emptyset$  then
5:     for all  $M_{t,k} \in \mathcal{M}_t$  do
6:        $O_{\text{new}} \leftarrow \Phi(\mathcal{G}, v_t, M_{t,k})$ 
7:        $\mathcal{O} \leftarrow \mathcal{O} \cup \{O_{\text{new}}\}$ 
8:     end for
9:     continue
10:  end if
11:   $\mathcal{P} \leftarrow \emptyset$ 
12:  for all  $O_j \in \mathcal{O}$  do
13:     $\widehat{M}_{j,t} \leftarrow \Pi_t(O_j)$ 
14:    for all  $M_{t,k} \in \mathcal{M}_t$  do
15:       $q_{j,k} \leftarrow \text{IoU}(\widehat{M}_{j,t}, M_{t,k})$ 
16:      if  $q_{j,k} \geq \tau_{\text{IoU}}$  then
17:         $\mathcal{P} \leftarrow \mathcal{P} \cup \{(j, k, q_{j,k})\}$ 
18:      end if
19:    end for
20:  end for
21:  Sort  $\mathcal{P}$  by decreasing  $q_{j,k}$ 
22:   $\mathcal{O}_{\text{matched}} \leftarrow \emptyset, \mathcal{M}_{\text{matched}} \leftarrow \emptyset$ 
23:  for all  $(j, k, q_{j,k}) \in \mathcal{P}$  do
24:    if  $O_j \notin \mathcal{O}_{\text{matched}}$  and  $M_{t,k} \notin \mathcal{M}_{\text{matched}}$  then
25:       $\mathcal{O}_{\text{matched}} \leftarrow \mathcal{O}_{\text{matched}} \cup \{O_j\}$ 
26:       $\mathcal{M}_{\text{matched}} \leftarrow \mathcal{M}_{\text{matched}} \cup \{M_{t,k}\}$ 
27:    end if
28:  end for
29:  for all  $M_{t,k} \in \mathcal{M}_t \setminus \mathcal{M}_{\text{matched}}$  do
30:     $O_{\text{new}} \leftarrow \Phi(\mathcal{G}, v_t, M_{t,k})$ 
31:     $\mathcal{O} \leftarrow \mathcal{O} \cup \{O_{\text{new}}\}$ 
32:  end for
33: end for
34: return  $\mathcal{O}$ 

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2 2D segmentation results

Table 1. 2D segmentation and back-projected 3D segmentation results, reporting instance F1, merged Dice, and predicted-to-GT instance count. SAM automatic masks use the standard automatic mask generator. SAM + GT point and SAM + GT box use a point at the GT mask center and the GT bounding box, respectively. The final configuration 3D→2D evaluates our 3D instances after rendering them back into the annotated 2D views.

Method	F1@.30	F1@.50	Merged Dice	Pred./GT count
FoMo4Wheat-Large + Mask2Former	0.511	0.211	0.679	47.9/38.2
SAM automatic masks	0.204	0.185	0.102	89.2/38.2
SAM + GT point	0.387	0.350	0.390	10.9/38.2
SAM + GT box	0.925	0.728	0.815	36.8/38.2
Ours (3D→2D)	0.363	0.175	0.607	38.0/38.2

3 3DGS reconstruction

Table 2. 3DGS reconstruction results, reporting the view configuration, number of Gaussians, PSNR, SSIM, and LPIPS. The *densifySafe* configuration uses a densification-gradient threshold of 1×10^{-3} and densification until iteration 7,000; *densifyHigh* uses a threshold of 5×10^{-4} and densification until iteration 12,000.

Model	Views	Gaussians	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
densifySafe	700/100	375,398	27.95	0.868	0.170
densifyHigh	700/100	817,352	28.85	0.882	0.149
densifyHigh	800 all	820,619	28.89	0.885	0.148

4 Complete 3D segmentation comparison

Table 3. 3D instance segmentation results for the evaluated SA3D-GS and Wheat3DGS configurations, reporting predicted and matched instance counts, object-level F1, IoU, Dice, precision, and recall. “All” metrics use the full prediction set; “Match” metrics use matched target instances. Wheat3DGS uses FoMo masks directly (*FoMoDirect*) or as SAM box (*FoMoBox-SAM*) and box+mask prompts (*FoMoBoxMask-SAM*). SA3D-GS uses FoMo masks (*FoMoMask*) or GT masks (*GTMask*, reference setup). *Raw* denotes no cleanup; CC-light, CC-medium, and CC-strict apply increasing connected-component filtering.

Method	Pred.	Match	F1@.30	F1@.50	All IoU	All Dice	All P	All R	Match IoU	Match Dice	Match P	Match R
W3D FoMoDirect raw	35	6	0.218	0.073	0.202	0.293	0.238	0.405	0.371	0.526	0.427	0.718
W3D FoMoDirect CC-light	27	8	0.340	0.170	0.242	0.331	0.281	0.449	0.451	0.603	0.523	0.762
W3D FoMoDirect CC-medium	23	8	0.372	0.186	0.235	0.318	0.267	0.433	0.454	0.607	0.514	0.809
W3D FoMoDirect CC-strict	16	7	0.389	0.222	0.206	0.275	0.227	0.377	0.406	0.540	0.448	0.729
W3D FoMoBox-SAM raw	39	8	0.271	0.102	0.260	0.378	0.273	0.674	0.412	0.573	0.422	0.932
W3D FoMoBox-SAM CC-light	28.4	11	0.458	0.125	0.304	0.423	0.329	0.664	0.488	0.649	0.518	0.926
W3D FoMoBox-SAM CC-medium	23	9	0.419	0.186	0.286	0.394	0.294	0.677	0.488	0.644	0.499	0.949
W3D FoMoBox-SAM CC-strict	22	9	0.429	0.190	0.283	0.388	0.291	0.657	0.489	0.645	0.500	0.949
W3D FoMoBoxMask-SAM raw	44	8	0.250	0.031	0.225	0.337	0.245	0.641	0.369	0.533	0.393	0.910
W3D FoMoBoxMask-SAM CC-light	31	9	0.353	0.078	0.265	0.378	0.290	0.637	0.444	0.609	0.486	0.902
W3D FoMoBoxMask-SAM CC-medium	26	9	0.391	0.087	0.264	0.378	0.288	0.651	0.444	0.609	0.484	0.901
W3D FoMoBoxMask-SAM CC-strict	23	9	0.419	0.140	0.276	0.390	0.300	0.651	0.461	0.624	0.502	0.901
SA3D-GS FoMoMask CC-light	53	10	0.274	0.082	0.326	0.468	0.339	0.851	0.472	0.637	0.485	0.957
SA3D-GS FoMoMask CC-medium	44	9	0.281	0.094	0.320	0.467	0.328	0.891	0.434	0.599	0.439	0.976
SA3D-GS FoMoMask CC-strict	38	10	0.345	0.103	0.313	0.445	0.322	0.803	0.477	0.643	0.484	0.966
SA3D-GS GTMask	19	16	0.842	0.737	0.533	0.650	0.557	0.794	0.673	0.803	0.694	0.956

5 Qualitative 3D segmentation examples

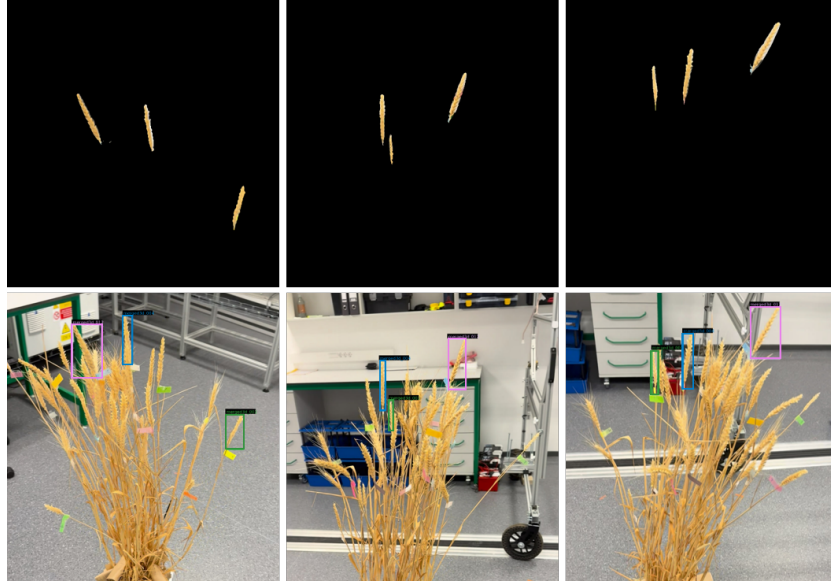


Fig. 1. Examples of 3D wheat ear segmentation rendered from three evaluation views. The upper row shows the selected Gaussian subsets; the lower row shows their projections over the corresponding RGB frames.

6 Per-ear morphometric measurements

Table 4. Physical reference measurements and estimated length, width, and volume for the measured wheat ears, together with measured mass.

Ear	Length (mm)			Width (mm)			Volume (mm ³)			Mass (g)
	Physical	Ours	Wheat3DGS	Physical	Ours	Wheat3DGS	CT	Ours	Wheat3DGS	
1	100.0	94.9	89.3	10.0	10.7	10.4	1904.7	1664.1	1615.4	0.56
2	100.0	105.5	90.3	10.0	14.8	9.4	2198.9	1626.8	1267.5	0.68
3	89.0	100.7	93.3	8.0	21.3	25.0	1252.7	1200.3	1032.6	0.29
5	81.0	79.0	91.9	8.0	10.7	12.6	1435.9	1293.4	1254.0	0.44
6	46.0	46.6	–	9.0	14.4	–	848.5	810.3	–	0.32
7	63.0	78.1	–	7.0	7.9	–	578.3	421.1	–	0.20
8	100.0	97.2	89.7	9.0	9.5	8.9	1906.5	1209.6	1094.1	0.44
9	84.0	83.2	99.0	10.0	9.8	21.8	1445.5	1572.8	1770.1	0.43